14.6 The chain rule

Suppose $z = f(x,y)$ and $x = g(t)$ and $y = h(t)$. Then $z = f(g(t), h(t))$ is a function of t. What is $\partial z / \partial t$?

The idea is that

$$\Delta x \approx \frac{\partial x}{\partial t}\Delta t, \qquad \Delta y \approx \frac{\partial y}{\partial t}\Delta t$$

$$\Delta z \approx \frac{\partial z}{\partial x}\Delta x + \frac{\partial z}{\partial y}\Delta y.$$

So

$$\Delta z \approx \frac{\partial z}{\partial x}\frac{\partial x}{\partial t}\Delta t + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}\Delta t$$

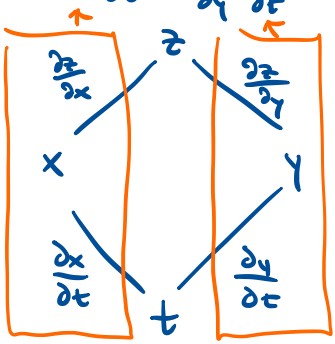
Then divide by Δt and "take the limit" to get

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t} \quad \leftarrow \text{Chain rule}$$

$$= f_x(g(t), h(t))\, g'(t) + f_y(g(t), h(t))\, h'(t).$$

Visually:

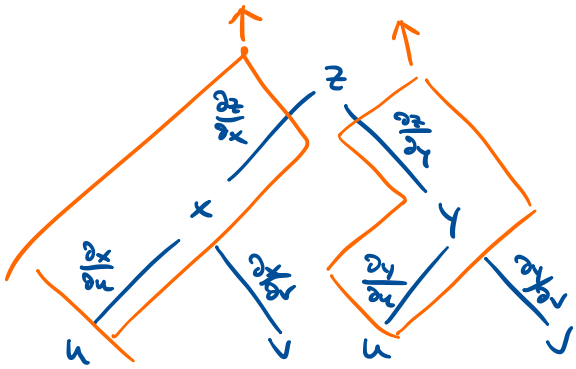
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$



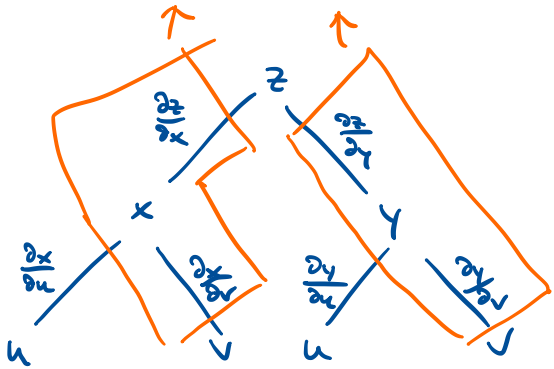
→ Do problems 1–3.

Now what if $z = f(x,y)$, and $x = g(u,v)$, $y = h(u,v)$? Then

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$



$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$



→ Do problem 4.